A nonvanishing, everywhere convergent local Rankin–Selberg integral for $\mathrm{GL}_{n+1} \times \mathrm{GL}_n$ with a unipotent shift

Abstract

Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$, and fix a nontrivial additive character $\psi : F \to \mathbb{C}^\times$ of conductor $\mathfrak{o}$. For any generic irreducible admissible representations Π of $\mathrm{GL}_{n+1}(F)$ and π of $\mathrm{GL}_n(F)$, we construct Whittaker functions $W \in \mathcal{W}(\Pi, \psi^{-1})$ and $V \in \mathcal{W}(\pi, \psi)$ such that the shifted Rankin–Selberg integral

$$\int_{N_n \backslash \mathrm{GL}_n(F)} W(\mathrm{diag}(g, 1)\, u_Q)\, V(g)\, |\det g|^{s-\frac{1}{2}}\, dg$$

is absolutely convergent for every $s \in \mathbb{C}$ and equals a nonzero constant (hence is finite and nonzero for all s). The key input is that the mirabolic restriction (Kirillov model) of a generic representation of $\mathrm{GL}_{n+1}(F)$ contains nonzero vectors compactly supported modulo N_{n+1}.

1 Notation

Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$, maximal ideal $\mathfrak{p}$, and normalized absolute value $|\cdot|$. For $r \geq 1$ put $G_r = \mathrm{GL}_r(F)$ and let $N_r \subset G_r$ denote the subgroup of upper-triangular unipotent matrices.

Fix $\psi : F \to \mathbb{C}^\times$ nontrivial of conductor $\mathfrak{o}$ (i.e. ψ is trivial on $\mathfrak{o}$ and nontrivial on $\mathfrak{p}^{-1}$), and use the standard generic character of N_r given by

$$\psi(n) = \psi\left(\sum_{i=1}^{r-1} n_{i,i+1}\right) \qquad (n \in N_r).$$

Let Π be a generic irreducible admissible representation of G_{n+1}, realized in its ψ^{-1}-Whittaker model $\mathcal{W}(\Pi, \psi^{-1})$. Let π be a generic irreducible admissible representation of G_n, realized in its ψ-Whittaker model $\mathcal{W}(\pi, \psi)$.

Let $\mathfrak{q}$ be the conductor ideal of π, choose a generator $Q \in F^\times$ of $\mathfrak{q}^{-1}$, and set

$$u_Q := I_{n+1} + Q\, E_{n,n+1} \in G_{n+1}.$$

(For the existence problem considered here, the particular choice of Q is irrelevant; any fixed $Q \in F^\times$ would do.)

For $W \in \mathcal{W}(\Pi, \psi^{-1})$ and $V \in \mathcal{W}(\pi, \psi)$ define

$$I(s; W, V) := \int_{N_n \backslash G_n} W\big(\mathrm{diag}(g, 1)\, u_Q\big)\, V(g)\, |\det g|^{s-\frac{1}{2}}\, dg, \tag{1}$$

where dg is a right Haar measure on G_n and the induced quotient measure is used on $N_n \backslash G_n$.

2 A Kirillov-model input

Let $P_r \subset G_r$ denote the mirabolic subgroup:

$$P_r = \{\, g \in G_r : \text{the last row of } g \text{ equals } (0, \ldots, 0, 1) \,\}.$$

Then P_r contains N_r.

For generic representations of G_r, restriction of Whittaker functions to P_r gives the (mirabolic) Kirillov model; it is a standard consequence of the Bernstein–Zelevinsky theory of derivatives that this Kirillov model contains a nonzero compactly supported (modulo N_r) subrepresentation. We isolate exactly what we need.

Proposition 1 (Compactly supported mirabolic Whittaker vectors)**.** *Let τ be a generic irreducible admissible representation of $G_r = \mathrm{GL}_r(F)$. Then there exists $W \in \mathcal{W}(\tau, \psi)$ such that $W|_{P_r}$ is compactly supported modulo N_r; equivalently,*

$$\{\, N_r p \in N_r \backslash P_r : W(p) \neq 0 \,\}$$

is compact. More precisely, the restriction map $\mathcal{W}(\tau, \psi) \to C^\infty(N_r \backslash P_r, \psi)$ has image containing the compact induction $\text{c-Ind}_{N_r}^{P_r}(\psi)$.

Remark 1. *Proposition* **??** *is a standard fact in the representation theory of $\mathrm{GL}_r(F)$, often phrased as: the Kirillov model of a generic representation contains a "compact Kirillov submodule". It can be deduced from the derivative filtration of $\tau|_{P_r}$ in Bernstein–Zelevinsky theory; see Bernstein–Zelevinsky [**?**] and, for an exposition in the language of Whittaker/Kirillov models, Matringe [**?**].*

3 Construction of W and V

Define the embedding (depending on Q)

$$j_Q : G_n \hookrightarrow P_{n+1}, \qquad j_Q(g) = \mathrm{diag}(g, 1)\, u_Q.$$

Lemma 1. *There exist a compact open subgroup $K \subset G_n$ and a Whittaker function $V \in \mathcal{W}(\pi, \psi)$ such that:*

1. *$K \subset \mathrm{GL}_n(\mathfrak{o})$ (hence $|\det k| = 1$ for $k \in K$ and ψ is trivial on $N_n \cap K$);*
2. *$V(k) = c$ for all $k \in K$, for some constant $c \in \mathbb{C}^\times$.*

Proof. Let $\Lambda : \pi \to \mathbb{C}$ be a nonzero (N_n, ψ)-equivariant functional. Choose $v_0 \in \pi$ with $\Lambda(v_0) \neq 0$. By smoothness, v_0 is fixed by some compact open subgroup $K_0 \subset G_n$. Put $K := K_0 \cap \mathrm{GL}_n(\mathfrak{o})$; then K is compact open, contained in $\mathrm{GL}_n(\mathfrak{o})$, and still fixes v_0.

Define $V(g) = \Lambda(\pi(g)v_0)$. Then for $k \in K$, $\pi(k)v_0 = v_0$, so $V(k) = \Lambda(v_0) =: c \neq 0$. □

Lemma 2. *Let $K \subset \mathrm{GL}_n(\mathfrak{o})$ be a compact open subgroup. There exists $W \in \mathcal{W}(\Pi, \psi^{-1})$ such that*

1. *$W(j_Q(k)) = 1$ for all $k \in K$;*
2. *$W(j_Q(g)) = 0$ unless $g \in N_n K$.*

Proof. Let $\Omega = j_Q(K) \subset P_{n+1}$. Define a function $f : P_{n+1} \to \mathbb{C}$ by

$$f(p) = \begin{cases} \psi^{-1}(u) & \text{if } p = u\,\omega \text{ with } u \in N_{n+1},\ \omega \in \Omega, \\ 0 & \text{if } p \notin N_{n+1}\Omega. \end{cases}$$

We claim that f is well-defined and lies in $\text{c-Ind}_{N_{n+1}}^{P_{n+1}}(\psi^{-1})$.

First, smoothness is clear and support is compact modulo N_{n+1} since Ω is compact. It remains to check well-definedness: suppose $u\omega = u'\omega'$ with $u, u' \in N_{n+1}$ and $\omega, \omega' \in \Omega$. Then $\omega'^{-1}\omega = u'^{-1}u \in N_{n+1} \cap \Omega\Omega^{-1}$. But

$$\Omega\Omega^{-1} = \big(\mathrm{diag}(K, 1)u_Q\big)\big(u_Q^{-1}\,\mathrm{diag}(K^{-1}, 1)\big) = \mathrm{diag}(K, 1),$$

so $N_{n+1} \cap \Omega\Omega^{-1} = N_{n+1} \cap \mathrm{diag}(K, 1) = \mathrm{diag}(N_n \cap K, 1)$. Since $K \subset \mathrm{GL}_n(\mathfrak{o})$ and ψ has conductor $\mathfrak{o}$, we have ψ trivial on $N_n \cap K$, hence ψ^{-1} trivial on $\mathrm{diag}(N_n \cap K, 1)$. Therefore $\psi^{-1}(u) = \psi^{-1}(u')$, and f is well-defined.

By Proposition **??** applied to $\tau = \Pi$ (with ψ replaced by ψ^{-1}), there exists $W \in \mathcal{W}(\Pi, \psi^{-1})$ whose restriction to P_{n+1} equals f.

By definition, $f(\omega) = 1$ for $\omega \in \Omega$, so $W(j_Q(k)) = 1$ for $k \in K$. Finally, if $W(j_Q(g)) \neq 0$, then $j_Q(g) \in N_{n+1}\Omega$. So $j_Q(g) = u\, j_Q(k)$ for some $u \in N_{n+1}$ and $k \in K$. Right-multiplying by u_Q^{-1} gives

$$\mathrm{diag}(g, 1) = u\ \mathrm{diag}(k, 1).$$

Thus $u = \mathrm{diag}(gk^{-1}, 1) \in N_{n+1}$, so $gk^{-1} \in N_n$ and $g \in N_n K$. □

4 Main theorem

Theorem 1. *In the setup above, for each fixed generic Π of $\mathrm{GL}_{n+1}(F)$ and each fixed generic π of $\mathrm{GL}_n(F)$, there exist $W \in \mathcal{W}(\Pi, \psi^{-1})$ and $V \in \mathcal{W}(\pi, \psi)$ such that the integral (**??**) is absolutely convergent for every $s \in \mathbb{C}$ and is a nonzero constant function of s. In particular, it is finite and nonzero for all $s \in \mathbb{C}$.*

Proof. Choose K and V as in Lemma **??**, so $V(k) = c \neq 0$ for all $k \in K$, and $K \subset \mathrm{GL}_n(\mathfrak{o})$. Choose W as in Lemma **??** for this K.

Lemma **??** implies that the integrand in (**??**) vanishes unless $g \in N_n K$, so the integral is supported on the compact subset $N_n \backslash N_n K \simeq (N_n \cap K)\backslash K$ of $N_n \backslash G_n$. Hence (**??**) converges absolutely for every $s \in \mathbb{C}$.

Write $g = nk$ with $n \in N_n$ and $k \in K$. Since V is a Whittaker function,

$$V(nk) = \psi(n)\, V(k) = \psi(n)\, c.$$

Also, since W is a ψ^{-1}-Whittaker function on G_{n+1} and $\mathrm{diag}(n, 1) \in N_{n+1}$,

$$W(\mathrm{diag}(nk, 1)u_Q) = W(\mathrm{diag}(n, 1)\,\mathrm{diag}(k, 1)u_Q) = \psi^{-1}(n)\, W(j_Q(k)) = \psi^{-1}(n) \cdot 1.$$

Therefore

$$W(\mathrm{diag}(g, 1)u_Q)\, V(g) = \psi^{-1}(n)\psi(n)\, c = c$$

for all $g \in N_n K$. Moreover, $|\det g| = |\det(nk)| = |\det k| = 1$ for $k \in K$. Thus the integrand equals the constant c on $N_n \backslash N_n K$ and vanishes outside, so

$$I(s; W, V) = c \cdot \mathrm{vol}(N_n \backslash N_n K),$$

a nonzero constant independent of s. □

Remark 2. *Theorem **??** answers the posed question in the affirmative for each fixed pair (Π, π) (and the given u_Q). The constructed W depends on π (through the choice of a compact open subgroup K fixing a suitable Whittaker vector of π) and on Q through the embedding j_Q. Demanding a single W (depending only on Π) that works uniformly for all π is a significantly stronger "test vector" problem of a different nature.*

References

[1] I. N. Bernstein and A. V. Zelevinsky, Induced representations of reductive p-adic groups. I, *Ann. Sci. Ecole Norm. Sup. (4)* **10** (1977), 441–472.

[2] N. Matringe, Essential Whittaker functions for $\mathrm{GL}(n)$, *Documenta Math.* **18** (2013), 1191–1214.